

Intern Name : Nidhi Rajpurohit

Campus-helpdesk Assignment

Task 1

```
nidhi@nidhi-Predator-PHN16-72:~$ minikube start
W0910 15:19:14.446790 17686 main.go:294] Unable to resolve the current Docker CLI context "default": context "default": context not found: open /home/nidhi/.docker/contexts/meta/37a8
eec1ce19687d132fe29051dca629d164e2c4958ba141d5f4133a33f0688f/meta.json: no such file or directory
W0910 15:19:14.446827 17686 main.go:295] Try running `docker context use default` to resolve the above error
🐳 minikube v1.39.0 on Ubuntu 24.04
🔧 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📶 Pulling base image v0.0.51 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.37.0 on containerd 2.3.4 ...
❗ Failing to connect to https://registry.k8s.io/ from both inside the minikube container and host machine
💡 To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: storage-provisioner, default-storageclass
🏁 Done! kubectcl is now configured to use "minikube" cluster and "default" namespace by default
nidhi@nidhi-Predator-PHN16-72:~$ minikube status
W0910 15:19:49.759984 21696 main.go:294] Unable to resolve the current Docker CLI context "default": context "default": context not found: open /home/nidhi/.docker/contexts/meta/37a8
eec1ce19687d132fe29051dca629d164e2c4958ba141d5f4133a33f0688f/meta.json: no such file or directory
W0910 15:19:49.760024 21696 main.go:295] Try running `docker context use default` to resolve the above error
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

nidhi@nidhi-Predator-PHN16-72:~$ kubectl create namespace campus-helpdesk
namespace/campus-helpdesk created
nidhi@nidhi-Predator-PHN16-72:~$ kubectl get namespaces
NAME                STATUS   AGE
campus-helpdesk     Active   15s
default             Active   4d4h
kube-node-lease     Active   4d4h
kube-public         Active   4d4h
kube-system         Active   4d4h
staging             Active   4d3h
nidhi@nidhi-Predator-PHN16-72:~$ kubectl config set-context --current --namespace=campus-helpdesk
Context "minikube" modified.
nidhi@nidhi-Predator-PHN16-72:~$ kubectl config view --minify --output 'jsonpath={..namespace}'; echo
campus-helpdesk
nidhi@nidhi-Predator-PHN16-72:~$ kubectl get pods
No resources found in campus-helpdesk namespace.
nidhi@nidhi-Predator-PHN16-72:~$
```

Task 2

```
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ nano persistent-volume.yaml
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ nano persistent-volume-claim.yaml
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl apply -f persistent-volume.yaml
persistentvolume/postgresql-pv created
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get pv
NAME                CAPACITY  ACCESS MODES  RECLAIM POLICY  STATUS    CLAIM                                STORAGECLASS  VOLUMEATTRIBUTESCLASS  REASON  AGE
postgresql-pv       2Gi       RWO           Retain          Available default/data-my-db-postgresql-0     standard      <unset>                7s
pvc-bc2960af-d4a7-47ec-a083-438d7144a64c  8Gi       RWO           Delete          Bound     default/data-my-db-postgresql-0     standard      <unset>                4d2h
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl apply -f persistent-volume-claim.yaml
persistentvolumeclaim/postgresql-pvc created
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get pv
NAME                CAPACITY  ACCESS MODES  RECLAIM POLICY  STATUS    CLAIM                                STORAGECLASS  VOLUMEATTRIBUTESCLASS  REASON  AGE
postgresql-pv       2Gi       RWO           Retain          Available default/data-my-db-postgresql-0     standard      <unset>                23s
pvc-6de4892e-4c93-4247-9e3a-566d23035b6d  2Gi       RWO           Delete          Bound     campus-helpdesk/postgresql-pvc      standard      <unset>                2s
pvc-bc2960af-d4a7-47ec-a083-438d7144a64c  8Gi       RWO           Delete          Bound     default/data-my-db-postgresql-0     standard      <unset>                4d2h
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get pvc
NAME                STATUS    VOLUME                                     CAPACITY  ACCESS MODES  STORAGECLASS  VOLUMEATTRIBUTESCLASS  AGE
postgresql-pvc      Bound     pvc-6de4892e-4c93-4247-9e3a-566d23035b6d  2Gi       RWO           standard      <unset>                13s
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$
```

Task 3

```
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ nano configmap.yaml
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl apply -f configmap.yaml
configmap/campus-helpdesk-config created
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get configmap
NAME          DATA   AGE
campus-helpdesk-config  5       17s
kube-root-ca.crt      1       15m
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl describe configmap campus-helpdesk-config
Name:         campus-helpdesk-config
Namespace:    campus-helpdesk
Labels:       <none>
Annotations:  <none>

Data
====
APP_ENV:
----
development

APP_NAME:
----
Campus Helpdesk

DB_HOST:
----
postgresql

DB_PORT:
----
5432

LOG_LEVEL:
----
info

BinaryData
====

Events: <none>
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$
```

Task 4

```
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ nano secret.yaml
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl apply -f secret.yaml
secret/campus-helpdesk-secret created
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get secret
NAME          TYPE      DATA   AGE
campus-helpdesk-secret  Opaque    3       10s
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl describe secret campus-helpdesk-secret
Name:         campus-helpdesk-secret
Namespace:    campus-helpdesk
Labels:       <none>
Annotations:  <none>

Type: Opaque

Data
====
POSTGRES_DB:      8 bytes
POSTGRES_PASSWORD: 11 bytes
POSTGRES_USER:    8 bytes
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$
```

Task 5

```
nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ nano postgresql-deployment.yaml
nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ kubectl apply -f postgresql-deployment.yaml
deployment.apps/postgresql created
nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
postgresql    0/1     1            0           6s
nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ kubectl get pods
NAME          READY   STATUS             RESTARTS   AGE
postgresql-747d7c5559-xdbnb  0/1     ContainerCreating   0          14s
nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ nano postgresql-service.yaml
nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ kubectl apply -f postgresql-service.yaml
service/postgresql created
nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ kubectl get svc
NAME          TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
postgresql    ClusterIP     10.100.211.233  <none>        5432/TCP   5s
nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ kubectl get pods
NAME          READY   STATUS             RESTARTS   AGE
postgresql-747d7c5559-xdbnb  0/1     ContainerCreating   0          68s
nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ kubectl get pods
NAME          READY   STATUS             RESTARTS   AGE
postgresql-747d7c5559-xdbnb  0/1     ContainerCreating   0          116s
nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ kubectl get pods
NAME          READY   STATUS             RESTARTS   AGE
postgresql-747d7c5559-xdbnb  0/1     ContainerCreating   0          2m28s
nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ kubectl describe pod -l app=postgresql
Name:          postgresql-747d7c5559-xdbnb
Namespace:     campus-helpdesk
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
Start Time:    Thu, 10 Sep 2026 15:59:04 +0530
Labels:        app=postgresql
               pod-template-hash=747d7c5559
Annotations:   <none>
Status:        Pending
IP:            <none>
IPs:           <none>
Controlled By: ReplicaSet/postgresql-747d7c5559
Containers:
  postgresql:
    Container ID:
    Image:        postgres:16
    Image ID:
    Port:         5432/TCP
    Host Port:    0/TCP
    State:        Waiting
      Reason:     ContainerCreating
    Ready:        False
    Restart Count: 0
    Readiness:    exec [sh -c pg_isready -U "$POSTGRES_USER" -d "$POSTGRES_DB"] delay=10s timeout=5s period=5s #success=1 #failure=6
    Environment Variables from:
```

```
      Campus-helpdesk-secret Secret Optional: false
Environment:               <none>
Mounts:
  /var/lib/postgresql/data from postgresql-storage (rw)
  /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-lz9wm (ro)
Conditions:
  Type          Status
  PodReadyToStartContainers True
  Initialized    True
  Ready          False
  ContainersReady False
  PodScheduled   True
Volumes:
  postgresql-storage:
    Type:          PersistentVolumeClaim (a reference to a PersistentVolumeClaim in the same namespace)
    ClaimName:     postgresql-pvc
    ReadOnly:      false
  kube-api-access-lz9wm:
    Type:          Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName:  kube-root-ca.crt
    Optional:      false
    DownwardAPI:   true
QoS Class:          BestEffort
Node-Selectors:     <none>
Tolerations:        node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                    node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type          Reason          Age          From          Message
  ----          -
  Normal        Scheduled       2m52s        default-scheduler Successfully assigned campus-helpdesk/postgresql-747d7c5559-xdbnb to minikube
  Normal        Pulling         2m53s        kubelet       spec.containers[postgresql]: Pulling image "postgres:16"
nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ kubectl describe pod -l app=postgresql
Name:          postgresql-747d7c5559-xdbnb
Namespace:     campus-helpdesk
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
Start Time:    Thu, 10 Sep 2026 15:59:04 +0530
Labels:        app=postgresql
               pod-template-hash=747d7c5559
Annotations:   <none>
Status:        Pending
IP:            <none>
IPs:           <none>
Controlled By: ReplicaSet/postgresql-747d7c5559
Containers:
  postgresql:
    Container ID:
    Image:        postgres:16
    Image ID:
```

```

PodScheduled          True
Volumes:
  postgresql-storage:
    Type:      PersistentVolumeClaim (a reference to a PersistentVolumeClaim in the same namespace)
    ClaimName: postgresql-pvc
    ReadOnly:  false
  kube-api-access-lz9wm:
    Type:      Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName: kube-root-ca.crt
    Optional:   false
    DownwardAPI: true
QoS Class:           BestEffort
Node-Selectors:      <none>
Tolerations:         node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                     node.kubernetes.io/unreachable:NoExecute op=Exists for 300s

Events:
  Type      Reason      Age   From      Message
  ----      -
Normal     Scheduled   3m3s  default-scheduler  Successfully assigned campus-helpdesk/postgresql-747d7c5559-xdbnb to minikube
Normal     Pulling     3m3s  kubelet     spec.containers[postgresql]: Pulling image "postgres:16"
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl logs -l app=postgresql
Error from server (BadRequest): container "postgresql" in pod "postgresql-747d7c5559-xdbnb" is waiting to start: ContainerCreating
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get pods -w
NAME                                READY   STATUS    RESTARTS   AGE
postgresql-747d7c5559-xdbnb         0/1     ContainerCreating  0          3m55s

^Cnidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get pods -w
NAME                                READY   STATUS    RESTARTS   AGE
postgresql-747d7c5559-xdbnb         0/1     ContainerCreating  0          5m55s
^[[A][A][A][A^Cnidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get svc
NAME      TYPE      CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
postgresql  ClusterIP  10.100.211.233  <none>        5432/TCP   5m19s
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
postgresql-747d7c5559-xdbnb         0/1     ContainerCreating  0          6m16s
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
postgresql-747d7c5559-xdbnb         0/1     ContainerCreating  0          7m
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get pods -w
NAME                                READY   STATUS    RESTARTS   AGE
postgresql-747d7c5559-xdbnb         0/1     ContainerCreating  0          7m29s
postgresql-747d7c5559-xdbnb         0/1     Running    0          7m48s
postgresql-747d7c5559-xdbnb         1/1     Running    0          7m58s
^Cnidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
postgresql-747d7c5559-xdbnb         1/1     Running    0          8m11s
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ █

```

Task 6

```

nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ find . -maxdepth 3 -type f | sort
./backend-deployment.yaml
./configmap.yaml
./persistent-volume-claim.yaml
./persistent-volume.yaml
./postgresql-deployment.yaml
./postgresql-service.yaml
./secret.yaml
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ nano backend-deployment.yaml
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ nano backend-service.yaml
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ nano backend-deployment.yaml
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl apply -f backend-deployment.yaml
deployment.apps/backend created
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl apply -f backend-service.yaml
service/backend created
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get deployment backend
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
backend   0/2     2            0           14s
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get pods -l app=backend
NAME                                READY   STATUS             RESTARTS   AGE
backend-69987dfdf-mzvxx             0/1     ErrImageNeverPull  0          21s
backend-69987dfdf-vz9bz             0/1     ErrImageNeverPull  0          21s
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get deployment backend -w
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
backend   0/2     2            0           36s
^Cnidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get pods -l app=backend -w
NAME                                READY   STATUS             RESTARTS   AGE
backend-69987dfdf-mzvxx             0/1     ErrImageNeverPull  0          4m59s
backend-69987dfdf-vz9bz             0/1     ErrImageNeverPull  0          4m59s
^Cnidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get service backend
NAME      TYPE      CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
backend   ClusterIP  10.101.105.170  <none>        5000/TCP   5m47s
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get endpoints backend
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointsSlice
NAME      ENDPOINTS   AGE
backend   5m56s
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl describe deployment backend
Name:
Namespace:      campus-helpdesk
CreationTimestamp: Thu, 10 Sep 2026 16:27:18 +0530
Labels:
  <none>
Annotations:    deployment.kubernetes.io/revision: 1
Selector:
  app=backend
Replicas:      2 desired | 2 total | 0 available | 2 unavailable
StrategyType:  RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=backend
  Containers:
    backend:

```

```
Host Port: 0/TCP
Limits:
  cpu: 500m
  memory: 256Mi
Requests:
  cpu: 100m
  memory: 128Mi
Liveness: http-get http://:3000/health delay=20s timeout=1s period=10s #success=1 #failure=3
Readiness: http-get http://:3000/health delay=10s timeout=1s period=5s #success=1 #failure=3
Environment Variables from:
  campus-helpdesk-config ConfigMap Optional: false
  campus-helpdesk-secret Secret Optional: false
Environment:
  <none>
Mounts:
  <none>
Volumes:
  <none>
Node-Selectors:
  <none>
Tolerations:
  <none>
Conditions:
  Type          Status Reason
  ----          -
  Available     False MinimumReplicasUnavailable
  Progressing   True ReplicaSetUpdated
OldReplicaSets: <none>
NewReplicaSet:  backend-69987dfdf (2/2 replicas created)
Events:
  Type          Reason          Age    From          Message
  ----          -
  Normal ScalingReplicaSet 6m13s deployment-controller Scaled up replica set backend-69987dfdf from 0 to 2
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl describe deployment backend
Name: backend
Namespace: campus-helpdesk
CreationTimestamp: Thu, 10 Sep 2026 16:27:18 +0530
Labels: <none>
Annotations: deployment.kubernetes.io/revision: 1
Selector: app=backend
Replicas: 2 desired | 2 updated | 2 total | 0 available | 2 unavailable
StrategyType: RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 25% max unavailable, 25% max surge
Pod Template:
  Labels: app=backend
  Containers:
    backend:
      Image: myapp:latest
      Port: 3000/TCP
      Host Port: 0/TCP
      Limits:
        cpu: 500m
        memory: 256Mi
      Requests:
        cpu: 100m
```

```
EnableUserlandProxy: true
UserlandProxyPath: /snap/docker/3579/bin/docker-proxy

nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ minikube profile list
W0910 16:37:33.639320 66559 main.go:294] Unable to resolve the current Docker CLI context "default": context "default": context not found: open /home/nidhi/.docker/contexts/meta/37a8
eec1ce19687d132fe29051dca629d164e2c4958ba141d5f4133a33f0688f/meta.json: no such file or directory
W0910 16:37:33.639341 66559 main.go:295] Try running 'docker context use default' to resolve the above error

PROFILE DRIVER RUNTIME IP VERSION STATUS NODES ACTIVE PROFILE ACTIVE KUBECONTEXT
minikube docker containerd 192.168.49.2 v1.37.0 OK 1 * *

nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ minikube image load myapp:latest --profile=minikube
W0910 16:37:58.784251 67012 main.go:294] Unable to resolve the current Docker CLI context "default": context "default": context not found: open /home/nidhi/.docker/contexts/meta/37a8
eec1ce19687d132fe29051dca629d164e2c4958ba141d5f4133a33f0688f/meta.json: no such file or directory
W0910 16:37:58.784289 67012 main.go:295] Try running 'docker context use default' to resolve the above error
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ docker save myapp:latest -o /tmp/myapp.tar
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ minikube image load /tmp/myapp.tar
W0910 16:38:44.707482 67792 main.go:294] Unable to resolve the current Docker CLI context "default": context "default": context not found: open /home/nidhi/.docker/contexts/meta/37a8
eec1ce19687d132fe29051dca629d164e2c4958ba141d5f4133a33f0688f/meta.json: no such file or directory
W0910 16:38:44.707505 67792 main.go:295] Try running 'docker context use default' to resolve the above error
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ minikube image ls | grep myapp
W0910 16:39:04.470433 67953 main.go:294] Unable to resolve the current Docker CLI context "default": context "default": context not found: open /home/nidhi/.docker/contexts/meta/37a8
eec1ce19687d132fe29051dca629d164e2c4958ba141d5f4133a33f0688f/meta.json: no such file or directory
W0910 16:39:04.470456 67953 main.go:295] Try running 'docker context use default' to resolve the above error
docker.io/library/myapp:latest
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl rollout restart deployment backend
deployment.apps/backend restarted
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get pods -l app=backend
NAME READY STATUS RESTARTS AGE
backend-65cbcc6b5b-nnzcd 0/1 Running 0 5s
backend-69987dfdf-mzvxx 1/1 Running 0 12m
backend-69987dfdf-vz9bz 1/1 Running 0 12m
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get pods -l app=backend
NAME READY STATUS RESTARTS AGE
backend-65cbcc6b5b-4nzb 1/1 Running 0 33s
backend-65cbcc6b5b-nnzcd 1/1 Running 0 44s
backend-69987dfdf-mzvxx 1/1 Terminating 0 13m
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get service backend
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
backend ClusterIP 10.101.105.170 <none> 5000/TCP 13m
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get endpoints backend
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME ENDPOINTS AGE
backend 10.244.0.11:3000,10.244.0.12:3000 13m
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get deployment backend
NAME READY UP-TO-DATE AVAILABLE AGE
backend 2/2 2 2 14m
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$
```


Task 7

```
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ nano frontend-deployment.yaml
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ nano frontend-service.yaml
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl apply -f frontend-deployment.yaml
deployment.apps/frontend created
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl apply -f frontend-service.yaml
service/frontend created
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get deployment frontend
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
frontend      2/2     2            2           13s
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get pods -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-db7ff7dcb-mlrfh   1/1     Running   0          17s
frontend-db7ff7dcb-nv92p   1/1     Running   0          17s
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get service frontend
NAME          TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
frontend      ClusterIP     10.98.60.248 <none>        80/TCP     16s
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get endpoints frontend
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME          ENDPOINTS          AGE
frontend      10.244.0.13:3000,10.244.0.14:3000 25s
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$
```

Task 8

```
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ minikub status
minikub: command not found
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ minikube status
W0912 20:13:40.142420 14058 main.go:294] Unable to resolve the current Docker CLI context "default": context "default": context not found: open /home/nidhi/.docker/contexts/meta/37a8eec1ce19687d132fe29051dca629d164e2c4958ba141d5f4133a33f0688f/meta.json: no such file or directory
W0912 20:13:40.142458 14058 main.go:295] Try running 'docker context use default' to resolve the above error
minikube
type: Control Plane
host: Stopped
kubelet: Stopped
apiserver: Stopped
kubeconfig: Stopped

nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ minikube start
W0912 20:13:45.946088 14087 main.go:294] Unable to resolve the current Docker CLI context "default": context "default": context not found: open /home/nidhi/.docker/contexts/meta/37a8eec1ce19687d132fe29051dca629d164e2c4958ba141d5f4133a33f0688f/meta.json: no such file or directory
W0912 20:13:45.946124 14087 main.go:295] Try running 'docker context use default' to resolve the above error
🐳 minikube v1.39.0 on Ubuntu 24.04
🔧 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📶 Pulling base image v0.0.51 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.37.0 on containerd 2.3.4 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: default-storageclass, storage-provisioner
👉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ minikube addons enable ingress
W0912 20:14:16.654678 19533 main.go:294] Unable to resolve the current Docker CLI context "default": context "default": context not found: open /home/nidhi/.docker/contexts/meta/37a8eec1ce19687d132fe29051dca629d164e2c4958ba141d5f4133a33f0688f/meta.json: no such file or directory
W0912 20:14:16.654724 19533 main.go:295] Try running 'docker context use default' to resolve the above error
💡 ingress is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWNERS
   ▪ Using image registry.k8s.io/ingress-nginx/controller:v1.15.1
   ▪ Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.9
   ▪ Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.9
🔍 Verifying ingress addon...
🌟 The 'ingress' addon is enabled
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get pods -n ingress-nginx
NAME          READY   STATUS    RESTARTS   AGE
ingress-nginx-admission-create-rfj4t   0/1     Completed   0          2m17s
ingress-nginx-admission-patch-klvmk     0/1     Completed   0          2m17s
ingress-nginx-controller-d7cd8c989-mztfn 0/1     Running     0          2m17s
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get pods -n ingress-nginx
NAME          READY   STATUS    RESTARTS   AGE
ingress-nginx-admission-create-rfj4t   0/1     Completed   0          2m29s
ingress-nginx-admission-patch-klvmk     0/1     Completed   0          2m29s
ingress-nginx-controller-d7cd8c989-mztfn 1/1     Running     0          2m29s
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ nano ingress.yaml
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl apply -f ingress.yaml
ingress.networking.k8s.io/campus-helpdesk-ingress created
```

```

ingress-nginx-controller-6f66c2b9-rlc7m 1/1 Running 0 2m59s
nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ nano ingress.yaml
nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ kubectl apply -f ingress.yaml
ingress.networking.k8s.io/campus-helpdesk-ingress created
nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ kubectl get ingress
No resources found in default namespace.
nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ kubectl get ingress -n campus-helpdesk
NAME                CLASS    HOSTS      ADDRESS      PORTS      AGE
campus-helpdesk-ingress  nginx   *          80           30s
nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ kubectl describe ingress campus-helpdesk-ingress -n campus-helpdesk
Name:                campus-helpdesk-ingress
Labels:              <none>
Namespace:           campus-helpdesk
Address:
Ingress Class:       nginx
Default backend:     <default>
Rules:
  Host      Path      Backends
  ----      -
  *         /api      backend:5000 (10.244.0.5:3000,10.244.0.2:3000)
             /         frontend:80 (10.244.0.4:3000,10.244.0.3:3000)
Annotations: <none>
Events:
  Type     Reason      Age   From      Message
  ----     -
  Normal   Sync        40s   nginx-ingress-controller   Scheduled for sync

```

```

nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ minikube ip
W0912 20:21:52.768199    28534 main.go:294] Unable to resolve the current Docker CLI context "default": context "default": context not found: open /home/nidhi/.docker/contexts/meta/37a8
eec1ce19687d132fe29051dca629d164e2c4958ba141d5f4133a33f0688f/meta.json: no such file or directory
W0912 20:21:52.768236    28534 main.go:295] Try running 'docker context use default' to resolve the above error
192.168.49.2
nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ curl http://$(minikube ip)/
W0912 20:21:57.163016    28765 main.go:294] Unable to resolve the current Docker CLI context "default": context "default": context not found: open /home/nidhi/.docker/contexts/meta/37a8
eec1ce19687d132fe29051dca629d164e2c4958ba141d5f4133a33f0688f/meta.json: no such file or directory
W0912 20:21:57.163055    28765 main.go:295] Try running 'docker context use default' to resolve the above error
Server is healthynidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ curl http://$(minikube ip)/api
W0912 20:22:00.308014    28998 main.go:294] Unable to resolve the current Docker CLI context "default": context "default": context not found: open /home/nidhi/.docker/contexts/meta/37a8
eec1ce19687d132fe29051dca629d164e2c4958ba141d5f4133a33f0688f/meta.json: no such file or directory
W0912 20:22:00.308058    28998 main.go:295] Try running 'docker context use default' to resolve the above error
Server is healthynidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$

```

Task 9

```

nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ kubectl get pods -n campus-helpdesk -l app=backend
NAME                READY   STATUS    RESTARTS   AGE
backend-65cbcc6b5b-4nzb  1/1     Running   1 (33m ago)  2d4h
backend-65cbcc6b5b-nnzc  1/1     Running   1 (33m ago)  2d4h
nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ kubectl exec -n campus-helpdesk backend-65cbcc6b5b-4nzb -- getent hosts postgresql
10.100.211.233 postgresql.campus-helpdesk.svc.cluster.local postgresql.campus-helpdesk.svc.cluster.local postgresql
nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$ kubectl exec -n campus-helpdesk backend-65cbcc6b5b-4nzb -- getent hosts frontend
10.98.60.248 frontend.campus-helpdesk.svc.cluster.local frontend.campus-helpdesk.svc.cluster.local frontend
nidhi@nidhi-Predator-PHN16-72: ~/campus-helpdesk$

```

Task 10

```
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get deployment backend -n campus-helpdesk
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
backend   2/2     2            2           2d4h
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get endpoints backend -n campus-helpdesk
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME      ENDPOINTS   AGE
backend   10.244.0.2:3000,10.244.0.5:3000  2d4h
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl scale deployment backend --replicas=0 -n campus-helpdesk
deployment.apps/backend scaled
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get pods -n campus-helpdesk -l app=backend
NAME      READY   STATUS    RESTARTS   AGE
backend-65cbcc6b5b-4nzb 1/1     Terminating 1 (40m ago) 2d4h
backend-65cbcc6b5b-nnzc 1/1     Terminating 1 (40m ago) 2d4h
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get endpoints backend -n campus-helpdesk
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME      ENDPOINTS   AGE
backend   <none>      2d4h
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ curl http://$(minikube ip)/
W0912 20:54:21.399342 51366 main.go:294] Unable to resolve the current Docker CLI context "default": context "default": context not found: open /home/nidhi/.docker/contexts/meta/37a8eec1ce19687d132fe29051dca629d164e2c4958ba141d5f4133a33f0688f/meta.json: no such file or directory
W0912 20:54:21.399367 51366 main.go:295] Try running 'docker context use default' to resolve the above error
Server is healthynidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl scale deployment backend --replicas=2 -n campus-helpdesk
deployment.apps/backend scaled
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get pods -n campus-helpdesk -l app=backend
NAME      READY   STATUS    RESTARTS   AGE
backend-65cbcc6b5b-jxsrh 1/1     Running    0           12s
backend-65cbcc6b5b-m6qxd 1/1     Running    0           12s
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get endpoints backend -n campus-helpdesk
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME      ENDPOINTS   AGE
backend   10.244.0.16:3000,10.244.0.17:3000  2d4h
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$
```

Task 11

```
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get pods -n campus-helpdesk -o wide
NAME      READY   STATUS    RESTARTS   AGE   IP           NODE       NOMINATED NODE   READINESS GATES
backend-65cbcc6b5b-jxsrh 1/1     Running    0           8m36s  10.244.0.16  minikube   <none>           <none>
backend-65cbcc6b5b-m6qxd 1/1     Running    0           8m36s  10.244.0.17  minikube   <none>           <none>
frontend-db7ff7dcb-mlrfh 1/1     Running    1 (49m ago) 2d4h   10.244.0.4   minikube   <none>           <none>
frontend-db7ff7dcb-nv92p 1/1     Running    1 (49m ago) 2d4h   10.244.0.3   minikube   <none>           <none>
postgresql-747d7c5559-xdbnb 1/1     Running    1 (49m ago) 2d5h   10.244.0.12  minikube   <none>           <none>
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get services -n campus-helpdesk
NAME      TYPE      CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
backend   ClusterIP  10.101.105.170 <none>        5000/TCP  2d4h
frontend   ClusterIP  10.98.60.248 <none>        80/TCP    2d4h
postgresql ClusterIP  10.100.211.233 <none>        5432/TCP  2d5h
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get pv
NAME      CAPACITY   ACCESS MODES   RECLAIM POLICY   STATUS   CLAIM              STORAGECLASS   VOLUMEATTRIBUTESCLASS   REASON   AGE
postgresql-pv 2Gi        RWO            Retain            Available  campus-helpdesk/postgresql-pvc  standard        <unset>                  2d5h
pvc-6de4892e-4c93-4247-9e3a-566d23035b6d 2Gi        RWO            Delete            Bound     default/data-my-db-postgresql-0  standard        <unset>                  2d5h
pvc-bc2960af-d4a7-47ec-a083-438d7144a64c 8Gi        RWO            Delete            Bound     default/data-my-db-postgresql-0  standard        <unset>                  6d8h
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get pvc -n campus-helpdesk
NAME      STATUS   VOLUME      CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS   AGE
postgresql-pvc  Bound    pvc-6de4892e-4c93-4247-9e3a-566d23035b6d 2Gi        RWO            standard        <unset>                  2d5h
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get configmap -n campus-helpdesk
NAME      DATA   AGE
campus-helpdesk-config 5       2d5h
kube-root-ca.crt      1       2d5h
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get secret -n campus-helpdesk
NAME      TYPE      DATA   AGE
campus-helpdesk-secret Opaque    3       2d5h
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$ kubectl get ingress -n campus-helpdesk
NAME      CLASS   HOSTS             ADDRESS     PORTS     AGE
campus-helpdesk-ingress nginx    *                 192.168.49.2 80        46m
nidhi@nidhi-Predator-PHN16-72:~/campus-helpdesk$
```